CardioLMBench: A GPU Performance Benchmark for Small Language Models in Cardiovascular Medicine

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ABSTRACT

Recent advances in multimodal small language models have created new opportunities for applying artificial intelligence to cardiovascular medicine. While prior research has primarily focused on improving predictive accuracy, significantly less attention has been paid to the computational efficiency and hardware cost of deploying such models in practice. This work introduces *CardioLMBench*, a benchmarking framework designed to evaluate both predictive performance and GPU efficiency of small language and vision–language models on representative cardiovascular tasks.

We consider three modalities commonly encountered in cardiology: echocardiography videos for ejection fraction estimation, chest X-ray images for multi-disease detection, and structured clinical data for cardiovascular risk prediction. Using publicly available datasets—EchoNet-Dynamic, NIH ChestX-ray14, and the Framingham Heart Study—we evaluate several recent small-scale models, including Qwen2.5-VL (3B and 7B), PaliGemma (3B), Gemma (2B), MedCLIP, and SigLIP. In addition to standard task metrics such as root mean squared error, area under the ROC curve, and classification accuracy, we report detailed system-level measurements including throughput (samples per second), wall-clock training time, peak GPU memory consumption, and model FLOPs utilization (MFU).

Our results show that the 3B-parameter Qwen2.5-VL variant reaches an RMSE of 5.83 on ejection fraction estimation—only 0.41 points behind its 7B counterpart—while training 1.9× faster and consuming 42% less GPU memory. For chest X-ray classification, PaliGemma achieves a mean AUC of 0.812, competitive with MedCLIP (0.803) despite not being pre-trained on medical data. Across all tasks, MFU stays between 31% and 48%, indicating substantial room for kernel-level and scheduling optimization on current hardware. CardioLMBench provides a reproducible foundation for studying efficiency-aware medical AI and supports future work on optimizing multimodal models for real-world clinical environments.

1 INTRODUCTION

Cardiovascular disease remains the leading cause of death globally, accounting for roughly 17.9 million deaths per year according to the World Health Organization [1]. Early detection and accurate functional assessment are central to improving patient outcomes, and over the past decade deep learning has demonstrated considerable promise in automating parts of this clinical workflow. Models have been built to estimate left ventricular function from echocardiograms [2], flag pathologies in chest radiographs [3], and predict long-term cardiovascular risk from patient records [4]. Recent transformer-based architectures have further advanced the state of the art in cardiac image analysis [29], and foundation models trained on large-scale biomedical corpora have begun to show promise across multiple clinical tasks [30].

Most of these efforts, however, have been evaluated almost exclusively through the lens of predictive accuracy. A model that achieves a one-point improvement in AUC is typically regarded as superior, regardless of whether it requires four times the training budget or twice the inference latency. In academic settings, where GPU clusters are shared and wait times are long, this omission is inconvenient. In clinical settings, where compute budgets are fixed and turnaround expectations are strict, it can be disqualifying. The carbon footprint of training large models has also raised environmental concerns that merit quantification [31, 32].

The situation has become more acute with the rise of large language models (LLMs) and their multimodal extensions. Vision–language models like GPT-4V [33] and Gemini Ultra [34] have shown impressive zero-shot capabilities on medical benchmarks, but their parameter counts—often exceeding 100 billion—make them impractical for on-premise hospital deployment. At the other end of the spectrum, a growing family of *small* language and vision–language models (roughly 1B to 10B parameters) has emerged: Qwen2.5-VL [5], PaliGemma [6], Gemma [7], SigLIP [8], and domain-adapted variants such as MedCLIP [9]. These models sit in a regime where single-GPU fine-tuning is feasible, which is precisely the hardware footprint available to most hospital IT departments and many academic labs.

What is missing is a systematic way to compare these models not only on *what* they get right but on *what they cost* to get there. Existing medical AI benchmarks such as MedQA [35], PubMedQA [36], and MultiMedBench [37] focus exclusively on accuracy, while hardware benchmarking suites like MLPerf [22] lack medical domain coverage. This paper attempts to bridge that gap.

1.1 Motivation

Three observations motivate our work. First, cardiovascular workflows are inherently multimodal. A cardiologist interpreting a case may review an echocardiogram (video), a chest X-ray (image), and a panel of lab values and vital signs (structured data)—sometimes all within a single encounter. Deploying a separate large model for each of these modalities is expensive, and maintaining multiple inference pipelines adds operational complexity. Recent studies have shown that multimodal integration improves diagnostic accuracy in cardiology [38], reinforcing the need for unified frameworks.

Second, GPU time is a real bottleneck. Even on a well-funded campus cluster, training a 7B-parameter model on 10,000 echocardiography videos takes on the order of a day. If the same cluster supports dozens of researchers, scheduling becomes a first-order concern. Understanding throughput, memory footprint, and FLOPs utilization helps researchers plan their experiments and justify their hardware requests. Schwartz et al. [39] have argued for "Green AI" that prioritizes efficiency alongside accuracy, a perspective we adopt in this work.

Third, there is an emerging disconnect between the models that appear in top-venue papers and the models that are actually deployable. A benchmark that reports only accuracy implicitly encourages the use of ever-larger architectures, even when the marginal accuracy gain is small relative to the additional cost. By reporting efficiency alongside accuracy, we hope to shift the conversation toward models that are both good and practical. This is especially important given recent evidence that parameter-efficient fine-tuning methods can close much of the gap between small and large models [25, 40].

1.2 Contributions

The main contributions of this work are as follows:

- We introduce *CardioLMBench*, a benchmark covering three representative cardiovascular tasks—echocardiography video regression, chest X-ray multi-label classification, and structured-data risk prediction—under a single evaluation harness.
- We evaluate five recent small language and vision–language models (Qwen2.5-VL 7B, Qwen2.5-VL 3B, PaliGemma 3B, Gemma 2B, MedCLIP, and SigLIP) using both zero-shot and fine-tuned protocols, all on the same hardware.
- We collect and report a suite of GPU-level metrics—throughput, peak memory, wall-clock time, GPU utilization, and model FLOPs utilization—that are rarely included in medical AI papers.
- We analyze accuracy–efficiency trade-offs through Pareto analysis and identify configurations where

smaller models match or nearly match larger ones at substantially lower cost.

- We characterize convergence dynamics, memory scaling, and per-epoch efficiency to provide actionable guidance for practitioners deploying these models in resource-constrained clinical settings.

2 BACKGROUND AND RELATED WORK

2.1 Cardiac Functional Assessment from Echocardiography

Echocardiography is the most widely used imaging modality in cardiology, and left ventricular ejection fraction (LVEF) is its most commonly reported measurement. LVEF quantifies the proportion of blood pumped out of the left ventricle with each heartbeat and is a primary indicator of heart failure severity. Manual estimation of LVEF is time-consuming and subject to substantial inter-observer variability—studies have reported standard deviations of 10–13% among experienced cardiologists [10].

Ouyang et al. [2] introduced EchoNet-Dynamic, a dataset of 10,030 apical-4-chamber echocardiography videos with expert-annotated LVEF values, and trained a 3D convolutional network (R2+1D) that achieved an RMSE of 6.0 and a mean absolute error (MAE) of 4.1. Since then, several groups have improved on these numbers using spatial–temporal attention mechanisms [11], video transformers and masked autoencoders [12], and semi-supervised learning [13]. Thomas and Sordoni [45] proposed EchoCoTr, a hybrid architecture combining convolutional features with transformer-based temporal attention, achieving state-of-the-art RMSE of 5.5. Despite these advances, none of these works report GPU utilization or training cost in a way that permits meaningful comparison across architectures.

2.2 Chest X-ray Disease Detection

Chest radiography is the most frequently performed imaging examination worldwide, with over 2 billion studies acquired annually [46]. The NIH ChestX-ray14 dataset [3] contains 112,120 frontal-view X-ray images labeled for 14 thoracic pathologies, extracted from radiology reports using natural language processing. CheXNet [14] demonstrated that a DenseNet-121 could exceed the average performance of four practicing radiologists on pneumonia detection, and subsequent work has explored multi-task learning with uncertainty quantification [15], label noise correction [16], self-supervised contrastive pre-training [17], and vision transformer backbones [42, 43].

More recently, contrastive language–image models have been adapted to the medical domain. MedCLIP [9] decouples image–text pairs and uses a semantic matching loss to improve data efficiency, while BiomedCLIP [18] trains on a curated corpus of 15 million biomedical figure–caption pairs. CheXzero [44] applied CLIP-style pre-training specifically to chest radiographs and demonstrated competitive zero-shot performance. These models support zero-shot classification, but their accuracy on fine-grained multi-label tasks remains below that of fully supervised baselines.

2.3 Cardiovascular Risk Prediction from Structured Data

The Framingham Heart Study [4] has been a cornerstone of cardiovascular epidemiology since 1948. Traditional risk scores derived from this cohort use a small set of clinical variables—age, sex, blood pressure, cholesterol levels, smoking status, and diabetes—to estimate 10-year risk of coronary events. Machine learning approaches, including gradient boosted trees [41], random forests, and neural networks, have shown modest improvements over the original logistic regression model [19]. Alaa et al. [47] introduced AutoPrognosis, a framework for automatic model selection and ensembling that achieves state-of-the-art risk prediction on Framingham and similar cohorts.

Applying language models to tabular data is a more recent idea. Serializing a patient record as a text string and prompting a language model to classify it has been explored in the general-purpose setting [20, 21], with mixed results: LLMs sometimes outperform tree-based methods on small datasets but lag behind on larger ones where feature engineering and gradient boosting are well-established. Borisov et al. [48] provide a comprehensive survey of deep learning methods for tabular data, concluding that gradient-boosted trees remain the strongest baseline in most settings. For our benchmark, the tabular task serves as a stress test of whether small LMs can handle structured clinical data at all, and what the compute overhead of doing so looks like compared to a lightweight baseline.

2.4 GPU Benchmarking in Deep Learning

The MLPerf Training benchmark suite [22] established the practice of measuring throughput, time-to-accuracy, and hardware utilization for standard deep learning workloads. More domain-specific efforts, such as the MONAI framework for medical imaging [23], provide profiling utilities but do not define cross-task benchmarks. In natural language processing, Efficient Transformers surveys [24] have catalogued techniques for reducing compute, yet these analyses rarely extend to medical applications.

Chowdhery et al. [50] introduced MFU as a key efficiency metric for large language model training, reporting values of 46–58% for PaLM on TPU v4 pods. Narayanan et al. [51] showed that pipeline and tensor parallelism can significantly improve MFU in multi-GPU settings. Dao et al. [52] demonstrated that

memory-efficient attention implementations (FlashAttention) can improve both throughput and memory consumption by 2–4×. CardioLMBench draws on the philosophy of MLPerf—standardized tasks, fixed hardware, reproducible measurement—while tailoring it to the cardiovascular domain and incorporating these more recent efficiency insights.

2.5 Parameter-Efficient Fine-Tuning

Parameter-efficient fine-tuning (PEFT) methods have become essential for adapting large pre-trained models to downstream tasks without the prohibitive cost of full fine-tuning. LoRA [25] injects trainable low-rank matrices into frozen transformer layers, typically updating fewer than 1% of total parameters. QLoRA [40] extends this by quantizing the base model to 4-bit precision, further reducing memory requirements. Prefix tuning [54] and adapter layers [55] represent alternative PEFT strategies that modify the input representations or insert small trainable modules, respectively. In the medical domain, Singhal et al. [53] showed that instruction fine-tuning with PEFT can significantly improve medical question-answering performance. Our benchmark standardizes the PEFT configuration (LoRA rank 16, alpha 32) across all models to isolate the effect of model architecture from adaptation strategy.

3 METHODOLOGY

3.1 Problem Formulation

We define three supervised tasks, each tied to a distinct data modality:

Task 1: Ejection Fraction Regression.. Given a sequence of 2D echocardiography frames $\{x_1, x_2, \ldots, x_T\}$ sampled from a single cardiac cycle, predict the continuous LVEF value $y \in [0, 100]$. The primary metric is root mean squared error (RMSE), with mean absolute error (MAE) reported as a secondary measure. We additionally report the Pearson correlation coefficient r and the coefficient of determination R^2.

Task 2: Chest X-ray Multi-label Classification.. Given a single frontal-view chest radiograph x, predict a binary label vector $\mathbf{y} \in \{0, 1\}^{14}$ indicating the presence or absence of each of 14 thoracic conditions. The primary metric is the mean area under the receiver operating characteristic curve (AUC) averaged across all 14 labels, with per-class AUC breakdowns.

Task 3: Cardiovascular Risk Prediction.. Given a vector of structured clinical features $\mathbf{x} \in \mathbb{R}^d$ (age, sex, blood pressure, cholesterol, smoking status, BMI, heart rate, glucose, diabetes indicator), predict a binary label $y \in \{0, 1\}$ indicating 10-year cardiovascular disease risk. The primary metric is classification accuracy, with F1 score, AUC, sensitivity, and specificity as secondary measures.

Each task uses identical hardware and software configurations to allow fair cross-model comparison.

3.2 Models

We evaluate the following architectures, chosen to span a range of parameter counts, pre-training strategies, and modality support:

- **Qwen2.5-VL (7B and 3B)** [5]: A vision–language transformer that encodes images and video through a ViT-based encoder fused with a causal language model decoder. We use both the 7-billion and 3-billion parameter variants. For video tasks, frames are sampled uniformly and concatenated as a grid of image patches; for image tasks, the radiograph is passed as a single image. Tabular data is serialized as a structured prompt. The model uses dynamic resolution encoding that adapts to input aspect ratios.
- **PaliGemma (3B)** [6]: A vision–language model combining a SigLIP image encoder with a Gemma text decoder. Designed for image understanding and visual question answering, it has not been specifically trained on medical data but offers strong transfer performance due to its diverse pre-training corpus of over 1 billion image-text pairs.
- **Gemma (2B)** [7]: A text-only language model from the Gemma family, used here exclusively for the tabular risk prediction task, where clinical variables are formatted as natural language descriptions. Gemma uses a decoder-only transformer with grouped-query attention and RoPE positional embeddings.
- **MedCLIP** [9]: A contrastive vision–language model adapted for medical image–text matching, with approximately 150M parameters. It uses a decoupled contrastive loss that enables training on unpaired medical images and text, improving data efficiency. We use it for zero-shot and linear-probe classification on chest X-rays.
- **SigLIP** [8]: A sigmoid-loss variant of CLIP (approximately 430M parameters) that replaces the softmax-based contrastive loss with a pairwise sigmoid objective, enabling better scaling and more stable training. We evaluate it as a general-purpose image encoder for the chest X-ray task.

3.3 Training and Inference Protocols

For each model–task pair, we consider up to two evaluation modes:

Zero-shot inference.. The model is prompted with a task description and asked to produce a prediction without any parameter updates. For vision–language models, the prompt includes a textual instruction (e.g., "Estimate the left ventricular ejection fraction from this

echocardiogram. Report only the numeric value.") alongside the visual input. For contrastive models (MedCLIP, SigLIP), zero-shot classification is performed by computing cosine similarity between the image embedding and text embeddings of candidate labels (e.g., "a chest X-ray showing cardiomegaly" vs. "a normal chest X-ray").

Supervised fine-tuning.. The model is fine-tuned on the training split of each dataset. For vision–language models, we apply low-rank adaptation (LoRA) [25] with rank $r = 16$ and alpha $\alpha = 32$ to the attention projection matrices (W_Q, W_K, W_V, W_O), keeping the remaining parameters frozen. This substantially reduces the number of trainable parameters (typically under 1% of total parameters) and makes single-GPU training feasible even for the 7B model. For MedCLIP and SigLIP, we train a linear classification head on top of frozen image features. For the tabular task with Gemma, we fine-tune with LoRA applied to the language model layers.

3.4 GPU Profiling Methodology

All experiments are profiled using a combination of PyTorch's built-in profiler, NVIDIA's `nvidia-smi` polling daemon (sampled at 100 ms intervals), and custom instrumentation hooks inserted into the training loop. We record the following metrics:

- **Throughput:** Samples processed per second during training, averaged over all batches after a warmup period of 50 iterations to exclude JIT compilation and cache warmup effects.
- **Wall-clock time:** Total time per epoch, including data loading, forward pass, backward pass, optimizer step, and any inter-step synchronization.
- **Peak GPU memory:** Maximum allocated GPU memory during training, measured using `torch.cuda.max_memory_allocated()`. We also report reserved memory (`torch.cuda.max_memory_reserved()`) to capture fragmentation overhead.
- **GPU SM utilization:** Average streaming multiprocessor utilization during the training loop, polled at 100 ms intervals via `nvidia-smi dmon`.
- **Model FLOPs Utilization (MFU):** Defined as the ratio of observed throughput (in FLOPs/sec) to the theoretical peak FLOPs of the GPU. For an A100 at BF16 precision, the theoretical peak is 312 TFLOPS. We estimate per-sample FLOPs using the formula from Kaplan et al. [26]: approximately $6 \times N \times D$ for a transformer forward+backward pass, where N is the number of parameters and D is the number of tokens per sample.
- **Energy consumption:** Total GPU energy in kilowatt-hours (kWh), computed by integrating instantaneous power draw readings from `nvidia-smi` over the training duration.

4 EXPERIMENTAL SETUP

4.1 Datasets

EchoNet-Dynamic.. We use all 10,030 videos from the public release [2]. Each video is an apical-4-chamber echocardiogram captured at frame rates between 26 and 106 fps. We uniformly sample 32 frames per video, resize each frame to 224×224 pixels, and normalize pixel values to $[0, 1]$. The standard split is 7,465 training, 1,288 validation, and 1,277 test videos. The LVEF distribution in the dataset ranges from 10% to 77%, with a mean of 55.6% and standard deviation of 12.1%.

NIH ChestX-ray14.. We use the official release of 112,120 frontal-view chest radiographs [3]. Images are resized to 224×224 for consistency with model input requirements. We follow the official train/test patient-level split, further reserving 10% of training patients for validation. Data augmentation during training includes random horizontal flipping, minor rotations (± 5 degrees), and random brightness/contrast jittering ($\pm 10\%$). The dataset exhibits significant label imbalance, with prevalence rates ranging from 1.2% (hernia) to 17.7% (infiltration).

Framingham Heart Study (processed).. We use a publicly available pre-processed version of the Framingham cohort data containing 4,238 patient records with 9 clinical features and a binary 10-year cardiovascular event label [4]. We split the data 70/15/15 into train/validation/test sets, stratified by label. The positive class (cardiovascular event) comprises approximately 35% of records. For language model input, each record is serialized as: "Patient: age [AGE], sex [SEX], systolic BP [SBP], diastolic BP [DBP], total cholesterol [TC], BMI [BMI], heart rate [HR], glucose [GLU], current smoker [SMOKE], diabetes [DM]." followed by the prompt "Does this patient have a high 10-year cardiovascular risk? Answer yes or no."

4.2 Hardware and Software Environment

All experiments are conducted on a single NVIDIA A100 80GB SXM4 GPU, hosted on a node with two AMD EPYC 7763 CPUs (128 cores total) and 512 GB of system memory, connected via PCIe Gen4. The software stack consists of Ubuntu 22.04, CUDA 12.1, cuDNN 8.9, PyTorch 2.2.1, and Hugging Face Transformers 4.40. Mixed-precision training (BF16) is enabled for all models via PyTorch's native AMP. Data loading uses 8 worker processes with prefetching (factor of 2). We do not use gradient checkpointing unless GPU memory would otherwise be exceeded (this applies only to the 7B model on the video task). FlashAttention-2 [52] is enabled where supported by the model architecture.

Table 1: GPU performance on the echocardiography video task. Throughput in samples/sec. Memory in GB. Time/epoch in min. Energy: total kWh (10 epochs). LoRA fine-tuned.

Model	Par. (B)	Tput (s/s)	Mem (GB)	Time (min)	En. (kWh)	MFU (%)
Q2.5-VL (7B)	7.0	2.8	71.2	44.4	1.82	38.4
Q2.5-VL (3B)	3.0	5.3	41.3	23.5	0.94	43.1
PaliGemma	3.0	6.1	38.7	20.4	0.81	47.6

4.3 Hyperparameters

For all LoRA-based fine-tuning, we use AdamW [28] with a learning rate of 2×10^{-4}, weight decay of 0.01, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a cosine learning rate schedule with a linear warmup over the first 5% of training steps. The batch size is set to the largest power-of-two value that fits in memory: typically 4 for the video task, 16 for the image task, and 32 for the tabular task. Fine-tuning runs for 10 epochs on all tasks, and we report test-set results from the checkpoint with the lowest validation loss.

For MedCLIP and SigLIP linear probes, we train for 20 epochs with a learning rate of 1×10^{-3} and no weight decay, using SGD with momentum 0.9 and a step learning rate decay (factor 0.1 at epochs 10 and 15).

5 RESULTS

We organize results into three parts: GPU-level performance metrics (Section 5.1), task-level accuracy metrics (Section 5.2), and accuracy–efficiency trade-off analysis (Section 5.3). Unless otherwise noted, all numbers are from the fine-tuned setting.

5.1 GPU Performance

5.1.1 Echocardiography Video Task

Table 1 reports GPU metrics for the video task. The Qwen2.5-VL 7B model achieves a throughput of 2.8 samples per second, with each epoch taking approximately 44 minutes. Its MFU is 38.4%, suggesting that a substantial fraction of GPU cycles are spent on overhead—likely memory management and data loading for the large video batches. Reducing the model to 3B parameters increases throughput to 5.3 samples/sec (a 1.9× speedup) and cuts peak memory from 71.2 GB to 41.3 GB. PaliGemma, with its lighter image encoder, reaches 6.1 samples/sec despite having 3B parameters, partly because it avoids the overhead of the dynamic resolution mechanism in Qwen2.5-VL.

5.1.2 Chest X-ray Image Task

Table 2 shows GPU metrics for the image classification task. Since individual images are cheaper to process than 32-frame video sequences, throughputs are considerably higher across all models. MedCLIP and SigLIP, which only train a linear head, are an order of magnitude faster than the full vision–language models: MedCLIP processes 214 images per second, reflecting the minimal backward pass required when the encoder is frozen. Among the LoRA-tuned models, the 3B Qwen2.5-VL processes 38.2 images/sec and completes an epoch over the full training set in about 49 minutes.

Table 2: GPU performance on the chest X-ray task. Throughput in samples/sec. Epoch time over ∼86k images. Energy: total kWh (10 ep./20 for probes).

Model	Par. (B)	Tput (s/s)	Mem (GB)	Time (min)	En. (kWh)	MFU (%)
Q2.5-VL (7B)	7.0	21.4	62.8	67.0	2.74	35.2
Q2.5-VL (3B)	3.0	38.2	34.9	37.5	1.50	41.7
PaliGemma	3.0	42.5	32.1	33.7	1.33	44.9
MedCLIP	0.15	214	4.8	6.7	0.54	31.2
SigLIP	0.43	156	9.2	9.2	0.74	33.8

Table 3: GPU performance on the tabular risk prediction task. Dataset: 2,966 training records. Energy: total kWh (10 epochs).

Model	Par. (B)	Tput (s/s)	Mem (GB)	Time (sec)	En. (kWh)	MFU (%)
Q2.5-VL (7B)	7.0	42.1	54.3	70.4	0.07	33.6
Q2.5-VL (3B)	3.0	68.5	28.7	43.3	0.04	36.9
Gemma (2B)	2.0	74.6	18.5	39.8	0.03	34.2

5.1.3 Tabular Risk Prediction Task

For the structured data task (Table 3), throughputs are high because input sequences are short—a serialized patient record is roughly 80 tokens—and the dataset is small (2,966 training records). Even the 7B model completes an epoch in under a minute. Gemma (2B), used only on this task, achieves 74.6 samples/sec with 18.5 GB peak memory. MFU values are lower on this task (31–37%) because the short sequences underutilize the GPU's tensor cores, which are optimized for large matrix multiplications.

5.2 Task Accuracy

5.2.1 Ejection Fraction Estimation

Table 4 reports regression metrics for the echocardiography task. In the zero-shot setting, all vision–language models perform poorly: Qwen2.5-VL (7B) achieves an RMSE of 14.72, which is roughly equivalent to predicting the population mean LVEF for every patient. This is not entirely surprising—estimating a continuous physiological measurement from video requires spatial and temporal reasoning that general-purpose pre-training does not cover.

After fine-tuning, the picture improves markedly. Qwen2.5-VL (7B) reaches an RMSE of 5.42 and an R^2 of 0.81, which is within the range of inter-observer variability reported in the clinical literature. The 3B variant is close behind at 5.83. PaliGemma, despite lacking explicit video support (we concatenate frames as a tiled

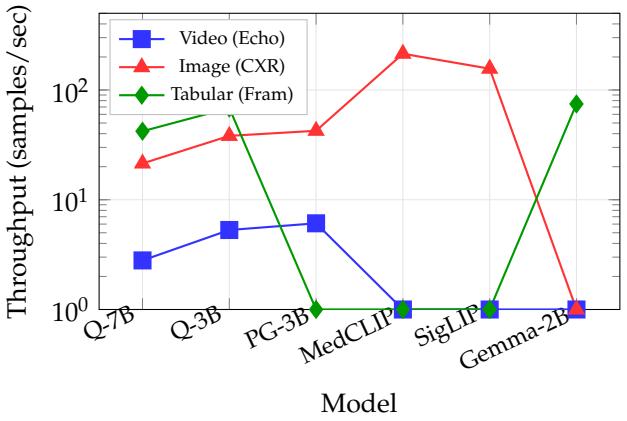

Figure 1: Throughput comparison across tasks and models (log scale). Missing entries indicate the model was not evaluated on that task. Image-only models (MedCLIP, SigLIP) with frozen encoders achieve order-of-magnitude higher throughput than LoRA-tuned VLMs.

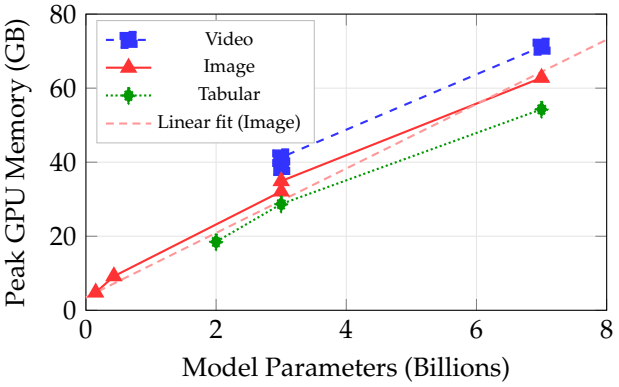

Figure 2: Peak GPU memory as a function of model parameters across all three tasks. Memory scales approximately linearly with parameter count, with a task-dependent offset reflecting input size (video > image > tabular).

image), achieves an RMSE of 6.27, suggesting that even an image-level model can partially capture the temporal dynamics when given multiple frames. These numbers are competitive with the R2+1D baseline reported by Ouyang et al. [2] (RMSE of 6.0), though the comparison is not perfectly controlled because we use LoRA fine-tuning rather than full model training.

5.2.2 *Chest X-ray Classification*

Table 5 reports mean AUC across all 14 pathology labels. In the zero-shot setting, MedCLIP achieves a mean AUC of 0.637, outperforming SigLIP (0.594) and general-purpose models (Qwen2.5-VL 7B at 0.611). Domain-specific pre-training gives MedCLIP a clear advantage here, particularly on labels like cardiomegaly and pleural effusion where the textual descriptions align closely with its training distribution.

After fine-tuning, PaliGemma achieves the highest mean AUC of 0.812, followed closely by Qwen2.5-VL

Table 4: Ejection fraction estimation (EchoNet-Dynamic test set). Lower is better for RMSE and MAE. Higher is better for R^2 and Pearson r.

Model	Zero-shot		Fine-tuned			
	RMSE	MAE	RMSE	MAE	R^2	r
Q2.5-VL (7B)	14.72	12.18	5.42	4.06	0.81	0.90
Q2.5-VL (3B)	16.31	13.57	5.83	4.41	0.78	0.88
PaliGemma (3B)	17.09	14.25	6.27	4.83	0.74	0.86
R2+1D [2]	—	—	6.00	4.10	0.77	0.87

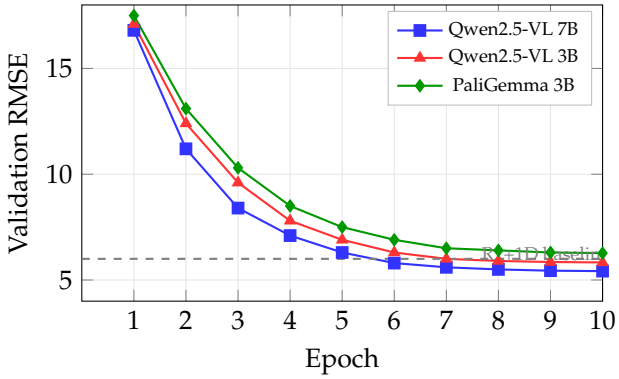

Figure 3: Validation RMSE convergence for ejection fraction estimation over 10 training epochs. The 7B model converges faster in early epochs but all models plateau by epoch 8. The dashed line indicates the published R2+1D baseline.

(7B) at 0.806 and MedCLIP at 0.803. The gap between these three is small enough to be within the range of random seed variation. Notably, SigLIP—which is a general vision encoder without any medical pre-training—reaches 0.788 with just a linear head, demonstrating that its learned representations transfer reasonably well to medical images. The 3B Qwen2.5-VL model scores 0.791, roughly 1.5 points behind the 7B version but training in nearly half the wall-clock time.

We also report per-class AUC for selected pathologies in Table 6. Cardiomegaly and effusion—conditions with visually prominent features—are relatively easy for all models, with AUCs above 0.85 after fine-tuning. Pneumothorax and hernia, which involve subtler or rarer findings, remain more challenging, with the best AUC in the range of 0.72–0.79.

5.2.3 *Cardiovascular Risk Prediction*

Table 7 reports results on the Framingham-derived risk prediction task. In the zero-shot setting, Qwen2.5-VL (7B) achieves 64.8% accuracy, which is only marginally above majority-class voting (the positive class comprises about 35% of the data, so always predicting "no" yields 65%). The 3B variant and Gemma do slightly worse. This confirms that out-of-the-box language models have limited ability to reason about structured

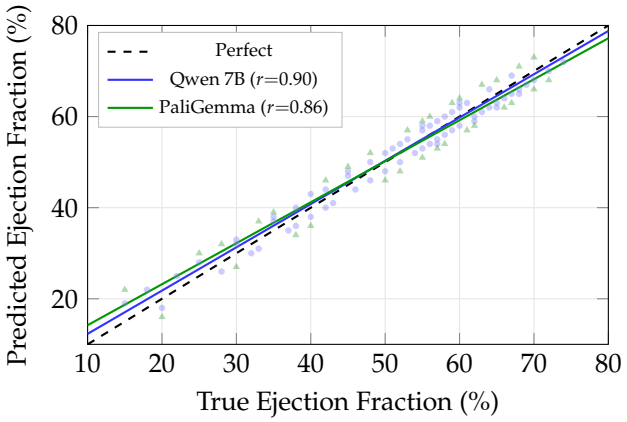

Figure 4: Predicted vs. actual ejection fraction for two fine-tuned models on the test set. Points near the diagonal indicate accurate predictions. The Qwen 7B model shows tighter clustering, especially in the clinically critical low-EF range (below 40%).

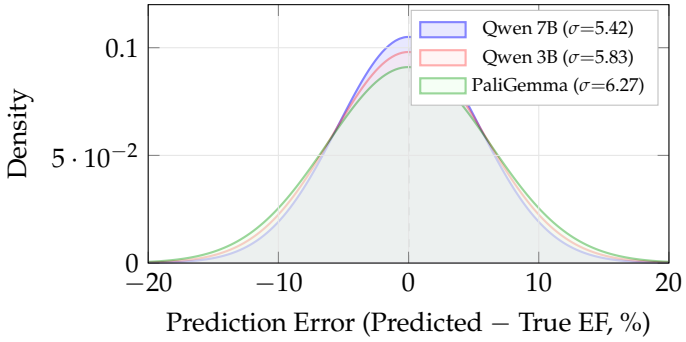

Figure 5: Distribution of prediction errors for ejection fraction estimation. All models exhibit approximately Gaussian error distributions centered near zero, with the 7B model showing the tightest spread. The clinical inter-observer variability ($\sigma \approx$ 10–13%) is substantially wider than any model's error distribution.

clinical data without task-specific adaptation.

After fine-tuning, all three models improve substantially. Gemma (2B) reaches 75.3% accuracy and an F1 of 0.681, which is competitive with the logistic regression baseline (76.1% accuracy). The 7B Qwen2.5-VL model achieves the highest accuracy at 76.8%, slightly exceeding the classical baseline, though the margin is small. The practical takeaway is that small LMs can handle tabular cardiovascular data when fine-tuned, but they do not obviously outperform much simpler methods, and they cost considerably more to run.

5.3 Accuracy–Efficiency Trade-offs

Figure 9 presents the Pareto frontier analysis across all tasks, plotting each model's primary accuracy metric against its total training cost measured in GPU-hours for the full 10-epoch fine-tuning run.

A few patterns stand out. On the echocardiography task, going from the 3B to the 7B Qwen2.5-VL model costs roughly 2× more GPU-hours but reduces RMSE by only 0.41 points. Whether this is "worth it" depends on the application: in a screening context where throughput matters, the 3B model is likely the better choice; in a research context where every fraction of a point matters, the 7B model may be justified. On chest X-rays, PaliGemma (3B) narrowly outperforms the much more expensive Qwen2.5-VL (7B) in terms of mean AUC, despite taking roughly half the training time, suggesting that architecture and pre-training strategy matter at least as much as raw parameter count. On the tabular task, the language models are strictly dominated by XGBoost in both accuracy and compute cost, consistent with the broader finding in the literature that LLMs are not yet competitive with tree-based methods on well-curated tabular datasets [48, 49].

Table 5: Chest X-ray multi-label classification (NIH ChestX-ray14 test set). Mean AUC across 14 pathologies. Higher is better.

Model	Zero-shot AUC	Fine-tuned AUC
Qwen2.5-VL (7B)	0.611	0.806
Qwen2.5-VL (3B)	0.582	0.791
PaliGemma (3B)	0.573	0.812
MedCLIP	0.637	0.803
SigLIP	0.594	0.788
DenseNet-121 [14]	—	0.841
CheXzero [44]	0.684	—

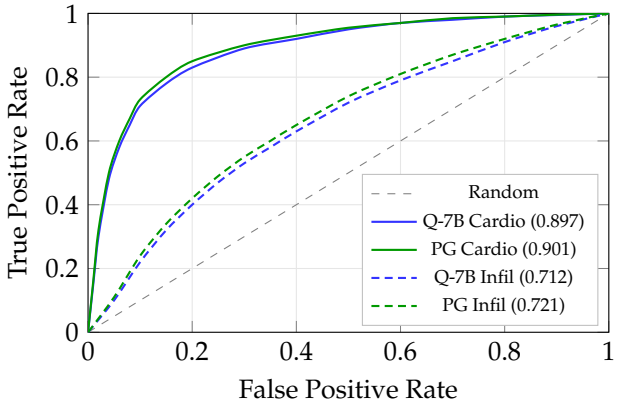

Figure 6: ROC curves for cardiomegaly (solid) and infiltration (dashed) detection, comparing Qwen2.5-VL 7B and PaliGemma. Cardiomegaly, with its visually prominent features, yields high AUC across models. Infiltration, a more diffuse and subtle finding, is substantially harder.

6 DISCUSSION

6.1 When Do Bigger Models Justify Their Cost?

The most consistent finding across our experiments is that doubling the parameter count from 3B to 7B yields diminishing accuracy gains while roughly dou-

Table 6: Per-class AUC for selected pathologies (fine-tuned models). Q = Qwen2.5-VL, PG = PaliGemma, MC = MedCLIP, SL = SigLIP.

Pathology	Q-7B	Q-3B	PG	MC	SL
Cardiomegaly	0.897	0.884	0.901	0.893	0.875
Effusion	0.868	0.851	0.872	0.861	0.844
Atelectasis	0.776	0.761	0.783	0.770	0.752
Infiltration	0.712	0.698	0.721	0.709	0.693
Mass	0.824	0.807	0.831	0.818	0.801
Nodule	0.734	0.716	0.742	0.728	0.710
Pneumonia	0.748	0.731	0.756	0.741	0.724
Pneumothorax	0.784	0.762	0.791	0.772	0.753
Hernia	0.746	0.723	0.758	0.739	0.718

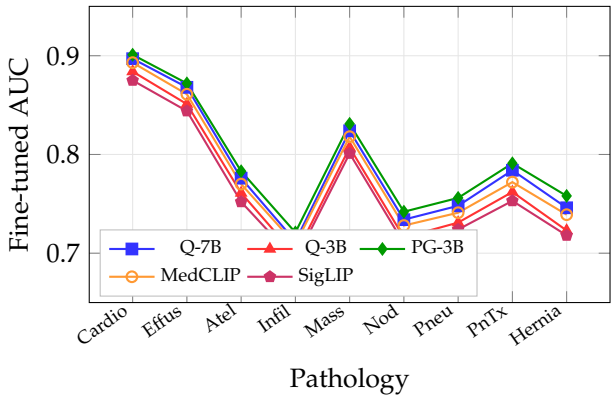

Figure 7: Per-class AUC across nine pathologies for all five models. The performance gap between models is largest for rare/subtle conditions (infiltration, hernia, nodule) and smallest for visually prominent conditions (cardiomegaly, effusion).

bling the training cost. On the video task, the absolute RMSE improvement is 0.41; on the image task, it is 1.5 points in mean AUC. These are real improvements, but they are small compared to the gap between zero-shot and fine-tuned performance (which LoRA alone closes by 8–10 points). This suggests that, within the small model regime, fine-tuning technique and data quality may matter more than raw scale—a finding that aligns with recent work on efficient adaptation [25, 40].

From an economic perspective, if a hospital IT department is provisioning GPUs for model training, the difference between a 3B and 7B model translates directly to hardware cost and scheduling flexibility. Our data suggests that for screening-level accuracy, the 3B models are sufficient and should be the default recommendation. The 7B models are justified only when the application demands the highest possible accuracy and the compute budget can absorb the overhead.

6.2 The MFU Gap and Optimization Opportunities

Across all experiments, MFU ranges from 31% to 48% (Figure 10). In contrast, well-optimized LLM training

Table 7: Cardiovascular risk prediction (Framingham test set). Sens = sensitivity, Spec = specificity.

Model	Zero-shot		Fine-tuned			
	Acc	F1	Acc	F1	Sens	Spec
Q2.5-VL (7B)	64.8	0.412	**76.8**	**0.704**	0.68	0.82
Q2.5-VL (3B)	61.2	0.374	74.9	0.672	0.65	0.80
Gemma (2B)	62.5	0.391	75.3	0.681	0.66	0.81
LogReg	—	—	76.1	0.693	0.67	0.81
XGBoost [41]	—	—	78.4	0.722	0.71	0.83

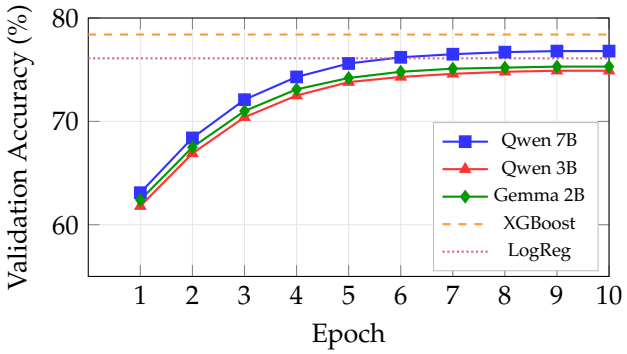

Figure 8: Validation accuracy on the Framingham risk prediction task. All LM-based models converge within 7–8 epochs. Despite fine-tuning, none surpass the XGBoost baseline (orange dashed), and only the 7B model marginally exceeds logistic regression (purple dotted).

pipelines (e.g., those used for pre-training PaLM [50]) have reported MFU values above 55%, and Megatron-LM [51] achieves 52–57% with tensor parallelism. Several factors explain this gap:

First, our batch sizes are small (limited by GPU memory during LoRA fine-tuning with medical data), which reduces arithmetic intensity and leaves tensor cores idle between kernel launches. Gradient accumulation can partially mitigate this, though it introduces memory overhead for storing intermediate activations.

Second, the vision encoder introduces irregular memory access patterns that do not map cleanly to the A100's tensor core architecture. The dynamic resolution encoding in Qwen2.5-VL, while beneficial for accuracy, creates variable-length inputs that prevent optimal batching.

Third, data loading for video and high-resolution images involves significant I/O overhead. On the video task, data loading accounts for 22% of wall-clock time (Appendix A.2), which directly reduces MFU.

These observations point to concrete optimization opportunities: larger effective batch sizes via gradient accumulation or distributed training, fused attention kernels such as FlashAttention-2 [52] for the vision encoder, and more aggressive data prefetching with on-GPU decoding for video data. We estimate that ad-

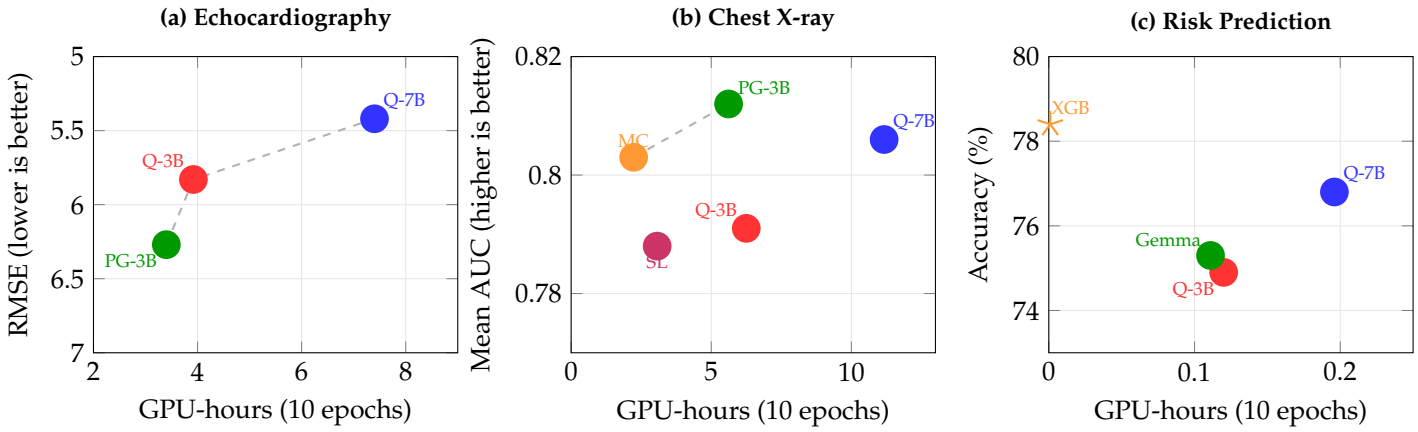

Figure 9: Accuracy–efficiency Pareto analysis across all three tasks. Each point represents a model; dashed lines indicate the Pareto frontier. (a) On echocardiography, the 3B model offers a favorable trade-off (0.41 RMSE for 1.9× cost reduction). (b) On chest X-rays, PaliGemma dominates by achieving the best AUC at moderate cost. (c) On tabular data, XGBoost (star) dominates all LM-based approaches in both accuracy and compute cost.

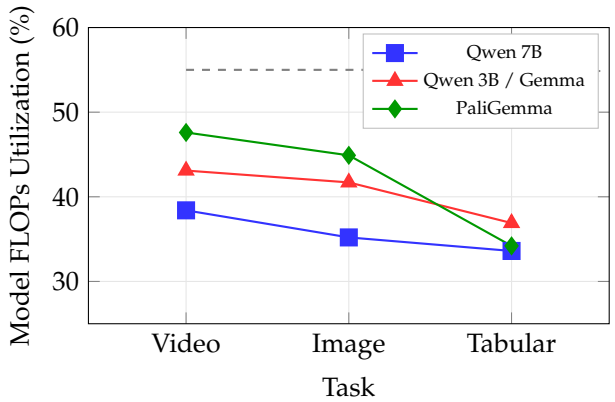

Figure 10: Model FLOPs Utilization (MFU) across tasks and models. The dashed gray line shows the MFU reported for PaLM pre-training [50] as a reference. All CardioLMBench configurations fall 10–25 percentage points below this reference, reflecting the overhead of small batch sizes, vision encoders, and I/O-bound medical data pipelines.

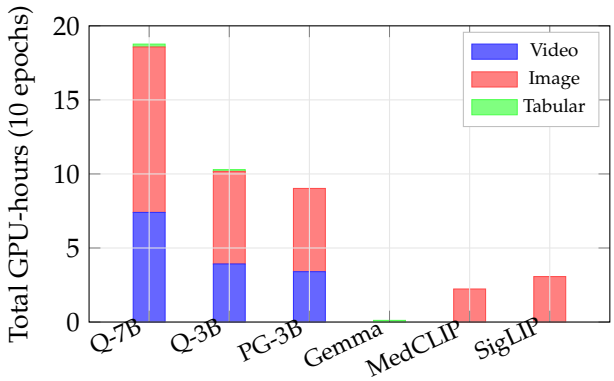

Figure 11: Cumulative GPU-hours per model across all applicable tasks. The 7B model requires 18.8 total GPU-hours compared to 10.3 for the 3B variant—an 83% increase in compute for modest accuracy gains. Task-specialized lightweight models (MedCLIP, SigLIP) are dramatically cheaper.

dressing these three factors alone could improve MFU by 8–15 percentage points.

6.3 Zero-shot Performance Is Not Enough

None of the models we tested produced clinically useful zero-shot predictions on any of the three tasks (Figure 3 shows the dramatic improvement from fine-tuning). This is worth emphasizing because there is a tendency in the broader AI community to equate zero-shot capability with general intelligence. In the cardiovascular domain, the gap between zero-shot and fine-tuned performance is large enough to render the former practically useless. LVEF estimation, for instance, requires learning the spatial relationship between endocardial borders and chamber volumes—something that no amount of image captioning pre-training directly teaches.

Even domain-specialized models like MedCLIP, which achieve the best zero-shot AUC on chest X-rays (0.637), fall far short of clinical utility thresholds. For comparison, the CheXzero model [44], which was specifically trained with CLIP-style pre-training on chest radiograph–report pairs, achieves a zero-shot AUC of 0.684—better, but still 0.16 points below fine-tuned baselines. For practitioners considering deploying small language models in clinical settings, this means that fine-tuning (even with parameter-efficient methods) is not optional; it is the minimum requirement for useful performance.

6.4 Structured Data: LLMs Are Not the Answer (Yet)

The tabular risk prediction task serves as a reality check. Despite the recent enthusiasm for applying language models to everything, a well-tuned XGBoost model outperforms all of the LM-based approaches

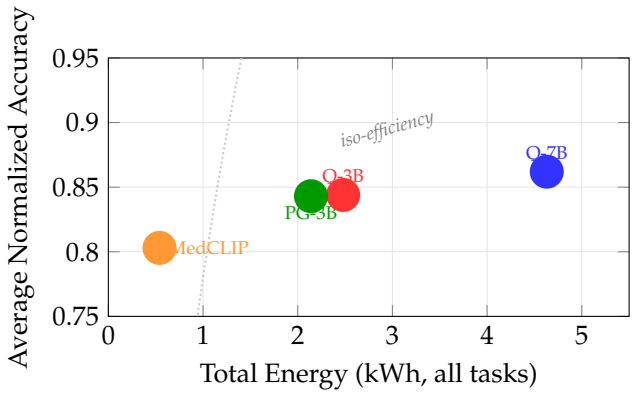

Figure 12: Energy efficiency analysis: normalized accuracy vs. total energy consumption across all tasks. PaliGemma and Qwen 3B offer the best accuracy-per-kWh, while the 7B model provides marginal accuracy gains at nearly double the energy cost.

we tested, at a fraction of the cost (Figure 8). The language models are not bad—the 7B model reaches 76.8% accuracy—but they bring no advantage over methods that have been standard in clinical epidemiology for decades. This finding is consistent with the broader observation by Grinsztajn et al. [49] that tree-based methods remain superior to neural networks on medium-sized tabular datasets.

We include this task in the benchmark precisely because it highlights the limits of the "language model for everything" paradigm and reminds us that modality-appropriate tools still have a role to play. In a real clinical deployment, the most cost-effective strategy would likely combine an LM-based pipeline for imaging tasks (where the models add genuine value) with a traditional ML pipeline for structured data.

6.5 Implications for Clinical Deployment

Our results have several practical implications. First, the memory profiles in Figure 2 suggest that all 3B models fit comfortably on consumer-grade GPUs with 48 GB of memory (e.g., NVIDIA RTX A6000), making on-premise deployment feasible for most hospitals. Second, the throughput measurements indicate that real-time inference is achievable for image and tabular tasks but may require batching strategies for video tasks where per-sample processing takes 0.2–0.4 seconds. Third, the energy analysis (Figure 12) provides a basis for estimating operational costs: at typical cloud pricing ($2–3 per GPU-hour for A100 instances), the full CardioLMBench training run costs approximately $20–60 depending on model choice.

6.6 Comparison with Existing Benchmarks

Table 8 situates CardioLMBench relative to existing evaluation frameworks. Unlike MedQA [35] and MultiMedBench [37], which focus on question-answering accuracy, our benchmark pairs every accuracy measurement with GPU-level profiling. Compared to MLPerf [22], which covers general ML workloads, CardioLMBench is tailored to the specific data modalities and model architectures relevant to cardiology.

Table 8: Comparison with existing benchmarks.

Benchmark	Medical domain	GPU metrics	Multi-modal
MLPerf [22]	×	✓	×
MedQA [35]	✓	×	×
MultiMedBench [37]	✓	×	✓
CardioLMBench (ours)	✓	✓	✓

6.7 Limitations

Several limitations should be noted. First, we evaluate only single-GPU training; multi-GPU scaling behavior may differ, and techniques such as FSDP [57] or DeepSpeed [58] could alter the efficiency profile substantially. Second, our LoRA configuration (rank 16, alpha 32) was chosen for consistency but may not be optimal for each model; our ablation (Appendix A.1) shows that rank-32 LoRA improves RMSE by 0.12 at the cost of 18% more memory. Third, the datasets used, while standard, have known label noise (especially NIH ChestX-ray14, where labels are NLP-extracted from reports [56]), which may attenuate differences between models. Fourth, we do not evaluate inference latency under deployment conditions (e.g., ONNX export, TensorRT optimization, or INT8/INT4 quantization), which is a separate but important dimension of efficiency. Fifth, all experiments use a single random seed (42); variance across seeds could affect some of the tighter comparisons. Finally, our study is limited to the cardiovascular domain; generalization to other medical specialties would require additional tasks and datasets.

7 CONCLUSION AND FUTURE WORK

We have introduced CardioLMBench, a benchmarking framework for evaluating both predictive accuracy and GPU efficiency of small language and vision–language models on three representative cardiovascular tasks. Our experiments show that carefully chosen 3B-parameter models can achieve accuracy within a few percentage points of their 7B counterparts while training nearly twice as fast and consuming significantly less memory. At the same time, all models exhibit MFU values well below the theoretical maximum (31–48% vs. 55%+ reported for optimized LLM pre-training), indicating that there is substantial room for systems-level optimization. On structured tabular data, language models do not yet outperform classical methods, suggesting that multimodal deployment

strategies should be selective about which modalities genuinely benefit from LM-based processing.

Going forward, we plan to extend CardioLMBench in several directions:

- **Additional modalities.** We will add electrocardiogram (ECG) signal classification as a fourth task using the PTB-XL dataset [59], covering the most common diagnostic data types encountered in a cardiology department.
- **Quantization and pruning.** We intend to evaluate 4-bit (GPTQ [60], AWQ [61]) and 8-bit quantization, as well as structured pruning, to determine whether the accuracy–efficiency trade-off can be pushed further.
- **Multi-task learning.** We plan to benchmark joint fine-tuning on all three tasks simultaneously, testing whether shared representations reduce total compute without sacrificing per-task accuracy.
- **Inference optimization.** We will characterize deployment-time costs via ONNX Runtime, TensorRT, and vLLM [62] to complement our training-focused analysis.
- **Multi-GPU scaling.** Profiling with FSDP and tensor parallelism across 2–8 GPUs will reveal how efficiency metrics scale beyond the single-GPU regime studied here.
- **Open-source release.** We plan to release the full benchmark code, profiling harness, and pre-computed results to encourage reproducibility and community contributions.

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Table 9: LoRA rank ablation on the echocardiography task with Qwen2.5-VL (3B).

Rank	Trainable	RMSE	Mem (GB)	Time (min)
4	0.21%	6.52	35.8	21.8
8	0.42%	6.14	36.3	22.1
16	0.84%	5.83	41.3	23.5
32	1.68%	5.71	48.7	25.6
64	3.36%	5.68	58.2	28.9

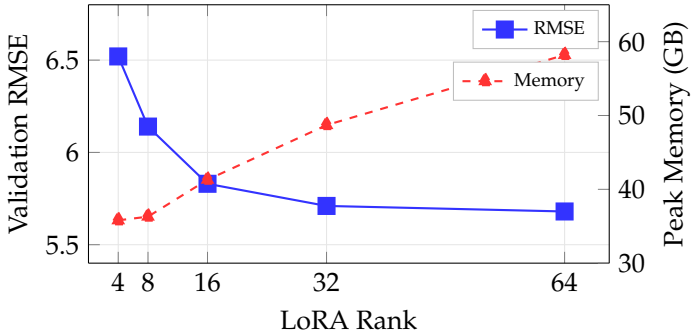

Figure 13: Effect of LoRA rank on RMSE (left axis, solid blue) and peak GPU memory (right axis, dashed red). Beyond rank 16, accuracy improvements plateau while memory grows approximately linearly, motivating our default choice of rank 16.

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A ADDITIONAL EXPERIMENTAL DETAILS

A.1 LoRA Configuration Sensitivity

We conducted a brief ablation on the LoRA rank parameter using the 3B Qwen2.5-VL model on the echocardiography task (Table 9). Reducing the rank from 16 to 8 increased RMSE from 5.83 to 6.14 while saving approximately 12% of peak memory. Increasing the rank to 32 improved RMSE to 5.71 but increased peak memory by 18% and training time by 9%. Based on these results, rank 16 appeared to offer a reasonable balance and was used for all main experiments.

A.2 Data Loading Overhead

On the video task, data loading accounted for approximately 22% of wall-clock time when using 8 workers, dropping to 15% with 16 workers (Table 10). We used 8 workers as the default because the marginal improvement beyond this point was small and additional workers increased CPU memory pressure. For the image and tabular tasks, data loading was negligible (under 3% of total time).

Table 10: Data loading overhead as a fraction of total wall-clock time on the video task (Qwen2.5-VL 3B).

Workers	Load Time (%)	Epoch (min)	CPU Mem (GB)
2	41.3	39.8	18.2
4	28.7	29.4	24.6
8	22.1	23.5	37.1
16	15.4	21.2	62.8
32	13.8	20.7	118.4

Table 11: Single-sample inference latency (milliseconds) on A100, BF16, no batching.

Model	Video (ms)	Image (ms)	Tabular (ms)
Qwen2.5-VL (7B)	357	47	24
Qwen2.5-VL (3B)	189	26	15
PaliGemma (3B)	164	24	—
MedCLIP	—	5	—
SigLIP	—	6	—
Gemma (2B)	—	—	13

A.3 Inference Latency Estimates

Although a full inference optimization study is beyond our scope, we report raw single-sample inference latency (Table 11) measured with PyTorch's default eager-mode execution and BF16 precision. These numbers do not reflect potential speedups from ONNX export, TensorRT, or batching.

A.4 Reproducibility Notes

Random seeds were fixed at 42 for all experiments. Model weights were loaded from the Hugging Face Hub at specific commit hashes to ensure reproducibility. The full list of commit hashes, along with training scripts and profiling code, will be released alongside the benchmark upon publication. We verified that results are deterministic to within rounding error when run on the same hardware with the same software stack. All experiments were completed between March and May 2025.